A Lightweight Autonomous Narrator Pipeline

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Abstract

The Lightweight Autonomous Narrator Pipeline demonstrates that long-form narrative generation, world hybridization, and stylistic steering can emerge from compact attractor architectures operating entirely on consumer-grade hardware. Rather than relying on large transformer models or extended-context inference, the pipeline combines a deterministic GLPG attractor, a lightweight narrator layer, a sinusoidally pretrained bias module, and a high-bandwidth entity-tracking vector into a modular IE×GR system capable of producing coherent, multi-chapter books without human steering.

Across three experimental axes—bias modulation, attractor distillation, and mixed-canon world hybridization—the system autonomously generates eighteen complete books. These include twelve Wonderland canon volumes produced under a wide range of synthetic mood intensities, three hybrid worlds formed from Wonderland/Kafka/Hyde with lemma blacklisting, and three high-coherence hybrid books generated through an iterative narrator pipeline with PRE-like chapter rollback. The results show stable mood control, predictable stylistic variation, strong chapter-level coherence, and robust world identity even under extreme modulation or mixed-canon conditions.

The experiments further reveal structural phenomena characteristic of self-sampling attractors, including manifold sharpening, entity amplification, and low-entropy convergence under distillation. Together, these findings position the narrator pipeline as a domain-bound demonstration of *architectural general intelligence*: a hybrid, inspectable, and reproducible AI ecosystem in which narrative competence emerges from modular structure rather than model scale.

1 Introduction

Long-form narrative generation is typically associated with large transformer models, extended-context inference, and datacenter-scale compute. Contemporary systems rely on probabilistic token prediction over millions of parameters, making narrative coherence, world consistency, and stylistic control emergent rather than architecturally guaranteed. In contrast, the present work develops a lightweight, modular, and fully inspectable alternative: a hybrid IE×GR narrator pipeline built on top of compact attractor architectures that operate entirely on consumer-grade hardware.

The GLPG attractor model provides a deterministic narrative motor in which lemma routes, pacing, canonical identity, and structural segmentation arise from the geometry of the learned manifold rather than from probabilistic sampling. The narrator pipeline augments this attractor with a lightweight moderation layer, a sinusoidally pretrained bias module functioning as a synthetic mood field, and a high-bandwidth entity-tracking vector that stabilizes world identity across long trajectories. Together, these components form a modular generative ecosystem in which narrative competence emerges from architectural structure rather than model scale.

This paper evaluates the pipeline across three experimental axes. First, bias modulation experiments demonstrate stable stylistic steering, predictable mood variation, and robust attractor behaviour under

extreme modulation intensities. Second, attractor distillation experiments reveal manifold sharpening, entity amplification, and low-entropy convergence when the system is trained on its own generated books. Third, mixed-canon hybridization experiments show how lemma blacklisting and narrator refinement can produce unified Alice-centric worlds from attractors trained on Wonderland, Kafka, and Hyde.

Across these experiments, the pipeline autonomously generates eighteen complete books: twelve Alice-canon volumes under varying bias intensities, three hybrid worlds produced through the draft narrator pipeline, and three high-coherence hybrid books generated through an iterative narrator pipeline with PRE-like chapter rollback. The results demonstrate strong chapter-level coherence, stable world identity, and controllable stylistic variation, even under mixed-canon or distilled conditions.

By combining deterministic attractor dynamics with modular cognitive layers, the Lightweight Autonomous Narrator Pipeline provides a domain-bound demonstration of *architectural general intelligence*: a hybrid, reproducible, and inspectable system in which narrative ability emerges from structural design rather than brute-force scale.

2 Related Work

Research on long-form narrative generation has traditionally focused on large transformer models and probabilistic sequence modeling. Systems such as GPT-style architectures, PaLM, LLaMA, and Mistral generate text through token-level next-word prediction over high-dimensional attention mechanisms. While these models can produce fluent prose, their narrative coherence, world consistency, and stylistic stability remain emergent properties of scale rather than architectural guarantees. Extended-context variants improve local continuity but do not provide deterministic structural control, canonical identity preservation, or interpretable world-model dynamics.

Story-specific LLM pipelines, including plot planners, outline generators, and chapter-level steering prompts, attempt to impose structure on transformer generation through external scaffolding. These approaches rely on prompt engineering, hierarchical planning, or multi-agent coordination, but they remain fundamentally probabilistic and lack stable internal world representations. Narrative drift, protagonist inconsistency, and stylistic collapse are common failure modes, especially in long-form settings.

Symbolic and procedural narrative engines, such as classical story grammars, planning-based systems, and rule-driven world simulators, offer deterministic structure but limited expressive richness. Their worlds are interpretable but not emergent; their narratives are coherent but not stylistically diverse. These systems lack the latent manifold dynamics that characterize modern generative models and therefore cannot produce the stylistic or thematic variability seen in LLM-based fiction.

Hybrid AI pipelines combine symbolic structure with neural generation, often through plot graphs, character sheets, or constraint-based steering. Although these systems improve controllability, they typically treat the neural component as a black-box generator and do not provide architectural integration between structure and emergence. They also lack mechanisms for deterministic routing, entity-aware world stabilization, or internal cognitive modulation.

Attractor-based models have been explored in cognitive science and dynamical systems research, but their application to narrative generation remains largely unexplored. Prior work on neural attractors focuses on memory retrieval, associative dynamics, or stable-state computation rather than long-form text generation. The GLPG architecture differs by embedding narrative canon, lemma routes, pacing, and structural segmentation directly into the attractor manifold, yielding deterministic world-model behaviour without attention mechanisms.

Finally, PRE-like rollback architectures appear in reinforcement learning, transactional planning, and multi-agent coordination, but they have not been applied to narrative generation. The iterative narrator pipeline introduced in this work integrates moderation, scoring, and continuity validation into a chapter-level PRE loop, providing structural guarantees that transformer-based systems cannot offer.

In summary, existing approaches either rely on large-scale probabilistic models with emergent structure or on symbolic systems with limited expressive richness. The Lightweight Autonomous Narrator Pipeline occupies a distinct position: a modular, attractor-driven architecture in which narrative coher-

ence, stylistic variation, and world identity arise from deterministic structure rather than model size or prompt engineering.

3 Lightweight Environment

A central claim of this work is that long-form narrative generation, autonomous world hybridization, and agentic narrator steering do not require large-scale transformer models or datacenter-class compute. The Lightweight Autonomous Narrator Pipeline operates entirely within a consumer-grade environment, using compact attractor architectures and a small 4-bit LLM to achieve capabilities typically associated with much larger systems.

3.1 Model Footprint

The GLPG attractor models used throughout this paper have a file size of only 60–70 MB. These models run entirely on CPU and require no GPU acceleration, no KV-cache, and no extended context windows. GLPG's deterministic transition map and fixed-size input vector (2000 floats) allow the attractor dynamics to function without the quadratic attention cost characteristic of transformer-based architectures.

The narrator layer employs a 4-bit quantized Mistral 7B v0.3 Instruct model (3.85 GB file size). This model runs comfortably on an 8 GB consumer GPU, a configuration widely available in mid-range laptops and desktops. No multi-GPU setup, high-bandwidth memory, or specialized inference hardware is required.

3.2 Hardware Configuration

All experiments, including autonomous draft generation, iterative narrator steering, bias modulation, and world hybridization, were executed on a single consumer laptop:

- Intel Core i7-14650HX CPU
- NVIDIA GeForce RTX 5070 Laptop GPU (8 GB VRAM)
- 16 GB system RAM

This configuration is representative of hardware accessible to private individuals, students, and independent researchers. It stands in contrast to contemporary long-context transformer systems, which often require 24–48 GB VRAM GPUs, high-end desktop workstations, or multi-hundred-gigabyte unified memory configurations such as those found in certain Mac Pro systems.

3.3 Architectural Efficiency

The lightweight nature of the pipeline is not a consequence of reduced capability, but of architectural design. GLPG provides an attractor-based narrative motor that encodes canon, lemma routes, pacing, and structural segmentation without relying on large probabilistic models. The narrator pipeline adds a compact agentic layer for bias modulation, world hybridization, and arc steering. Together, these components form an IE$\times$GR hybrid system in which architectural structure replaces brute-force scale.

3.4 Architecture Over Scale

The results presented in this paper demonstrate that a small model, when embedded in a structurally expressive attractor architecture and augmented with an agentic narrator layer, can achieve narrative coherence and world-model control typically associated with much larger LLMs. This is an architectural claim rather than a compute claim: the pipeline leverages structural determinism, attractor routing, and discrete narrator states to achieve capabilities that do not depend on model size or hardware scale.

In summary, the Lightweight Autonomous Narrator Pipeline shows that hybrid intelligence can emerge from compact models operating entirely on consumer-grade hardware. This challenges the prevailing assumption that long-form narrative generation and world-model reasoning require large transformer models or datacenter infrastructure.

4 Entity Tracking Vector

In addition to the structural, temporal, and syntactic signals described in [1], the input vector is extended with a lightweight *entity tracking* mechanism. This component provides a dynamic, sentence-level representation of the most salient noun and proper-name entities in the training corpus. Unlike narrative memory or token history, which encode multi-sentence structure, the entity tracker focuses exclusively on the evolving relevance of key entities across the generated document.

4.1 Top-Entity Selection

During preprocessing, all curriculum sentences are scanned for occurrences of grammar tokens belonging to the set

$$\langle \text{NOUN} \rangle,\ \langle \text{NOUN-PLURAL} \rangle,\ \langle \text{PROPN} \rangle.$$

Whenever such a grammar token is encountered, the following lemma token is guaranteed to be its semantic counterpart. For each lemma, a record is created, storing:

- the lemma string,
- its grammar category,
- the number of occurrences,
- the minimum and maximum line-fraction positions at which it appears.

Each entity receives a score combining frequency, positional spread, and grammar type. Entities are sorted by score, and the top K entities are selected. Each selected entity is assigned a fixed index in the entity-tracking vector.

4.2 Dynamic Tracking State

The **Entity Tracker** maintains a vector of length K, initialized to zeros. This vector evolves deterministically as each new sentence is generated. Before processing the sentence, the tracker applies an exponential decay:

$$s_i \leftarrow \alpha\, s_i,$$

where α is a configurable decay factor (e.g. $\alpha = 0.98$). This decay ensures that entity relevance fades smoothly over time unless reinforced by new occurrences.

The sentence is then scanned for grammar tokens in the tracker set. Whenever a tracked entity lemma appears immediately after such a grammar token, the corresponding vector entry is set to 1.0:

$$s_{\text{index}(e)} \leftarrow 1.0.$$

This produces a dynamic, sentence-level relevance signal: entities mentioned recently have high activation, while older entities gradually fade.

4.3 Advantages of the Entity Tracker

The entity-tracking vector offers several benefits:

- **Configurable bandwidth.** The number of tracked entities K can be chosen freely, allowing the model to operate with a small or large semantic footprint depending on the narrative domain.
- **Deterministic and interpretable.** Each entity corresponds to a single float in the input vector, making the mechanism transparent and easy to inspect.
- **Long-range semantic relevance.** The decay mechanism provides a lightweight memory of which entities have been active in recent sentences, without requiring recurrent networks or transformer-style attention.
- **Stable integration.** Because the tracker is independent of the attractor dynamics, it can be added without modifying the GLPG architecture. The recursive core simply receives an additional structured signal.

4.4 Role Within the Input Vector

The entity tracker complements the other components of the total input vector. While the current-sentence encoding provides micro-structure and token history offers multi-scale temporal modulation, the entity tracker supplies a semantic relevance profile that evolves smoothly across the narrative. This enables GLPG to maintain awareness of key entities over long trajectories, supporting more coherent world-model behavior without increasing model size or computational cost.

5 The New Input Vector

The Lightweight Autonomous Narrator Pipeline extends the GLPG input vector described in [1] with two additional structural signals: a normalized *paragraph line fraction* and a high-bandwidth *entity-tracking vector*. Together with updated configurations for token history, narrative memory, and current-sentence encoding, the new input vector reaches a total width of exactly 2000 floats. This represents a richer and more expressive contextual substrate while preserving the deterministic attractor dynamics of the GLPG core.

5.1 Paragraph Line Fraction

In addition to the global line fraction, which encodes the position of the current sentence within the entire generated document, a second positional scalar is introduced: the *paragraph line fraction*. This value resets to 0 at the start of each paragraph and increases linearly as new sentences are generated. The fraction is normalized by a fixed divider (e.g. 50), ensuring a stable and bounded positional signal:

$$\text{paragraph_fraction} = \frac{\text{line index within paragraph}}{50}.$$

This micro-positional cue provides the attractor system with fine-grained information about local paragraph progression, improving pacing, mid-scale segmentation, and structural rhythm.

5.2 Updated Token History

The token-history subsystem is redesigned to use five parallel history vectors, each with its own width and decay coefficient:

$$\text{history sizes} = [30, 40, 50, 60, 70], \qquad \text{history alphas} = [0.91, 0.93, 0.95, 0.97, 0.99].$$

Each history vector evolves according to

$$h \leftarrow \alpha h + (1 - \alpha)e,$$

where e is the current lemma embedding. The combined history signal contributes 250 floats to the input vector. These multi-scale traces provide layered temporal modulation, capturing both short-range and long-range lemma dynamics without requiring recurrent networks.

5.3 Reduced Narrative Memory

The narrative-memory module stores deterministic projections of the most recent sentences. In the new configuration, the memory depth is reduced from 20 to 8 sentences, each represented by a 32-float embedding:

$$8 \times 32 = 256 \text{ floats}.$$

This lighter memory preserves rhythmic smoothing and micro-callbacks while reducing bandwidth and improving generation speed.

5.4 Expanded Current-Sentence Encoding

The current-sentence subsystem is extended to include the last 15 grammar tokens and the last 15 lemma tokens. Each token contributes a 16-float embedding:

$$2 \times 15 \times 16 = 480.$$

Two additional scalars encode the normalized position within the sentence and a comma/punctuation indicator:

$$480 + 2 = 482 \text{ floats}.$$

This high-resolution snapshot of the ongoing sentence provides the strongest structural signal in the input vector, stabilizing syntax, pacing, and local narrative flow.

5.5 Entity Tracking

A new semantic subsystem tracks the dynamic relevance of the top 1010 entities extracted from the curriculum. Each entity corresponds to a single float in the input vector. Before processing each sentence, the tracker applies exponential decay:

$$s_i \leftarrow 0.98\, s_i.$$

Whenever a tracked entity appears in the sentence, its activation is set to 1.0. This produces a smooth, interpretable relevance profile across the narrative, enabling lightweight world-model coherence without increasing model complexity.

5.6 Line Fraction and Paragraph Line Fraction

Two normalized scalars encode global and local positional structure:

$$\text{line fraction} \in [0, 1], \qquad \text{paragraph line fraction} \in [0, 1].$$

These values anchor the attractor dynamics at both macro and micro scales.

5.7 Total Input Vector Size

The new input vector integrates all components into a single 2000-float encoding:

$$250\ (\text{token history}) + 256\ (\text{narrative memory}) + 482\ (\text{current}) + 1010\ (\text{entity tracking}) + 2\ (\text{line}) = 2000.$$

This expanded vector is a direct improvement over the 1307-float vector described in the previous GLPG paper. It provides richer structural, temporal, and semantic signals while maintaining full compatibility with the deterministic attractor architecture.

6 Generating a Suitable Title for the Book

While the GLPG attractor system autonomously generates chapter structure and chapter titles, the narrator pipeline includes an additional lightweight module that proposes suitable book titles based on the generated chapter headings. This component provides a simple but effective mechanism for selecting a coherent and stylistically appropriate title for each autonomously generated volume.

6.1 Title Generation from Chapter Headings

After the narrator pipeline has produced a complete book, including all chapter titles, the titler extracts these headings directly from the generated LaTeX file. Each chapter title functions as a high-level semantic summary of its corresponding chapter, and the ordered sequence of headings provides a compact representation of the book's thematic and structural identity.

Using this sequence, the titler constructs a concise semantic profile of the entire volume. No additional metadata, stylistic hints, or narrative prompts are introduced. The titler then invokes a lightweight LLM prompt that operates *exclusively* on the chapter-title list. The model receives only the ordered headings and is asked to synthesize a suitable book title from the structural signals already present in the attractor-generated draft.

This design preserves the autonomy of the generative pipeline. The LLM does not influence the book's content, steer its narrative, or impose external stylistic bias. It merely compresses the chapter-level structure into a human-readable title that reflects the book's thematic cohesion and narrative tone.

The titler produces multiple candidate titles—typically five—and the human operator selects the most appropriate one. This selection step is intentionally minimal: no rewriting or editing is required. The operator simply chooses the title that best captures the identity of the autonomously generated volume.

6.2 Lightweight Human-in-the-Loop Selection

Although the narrator pipeline is fully autonomous, the final book title is selected by a human operator. This selection step is intentionally lightweight: the operator chooses one title from a small set of candidates. No editing, rewriting, or structural modification is required.

This design preserves the autonomy of the generative system while ensuring that the final published title aligns with human expectations of clarity, tone, and suitability.

6.3 Role Within the Pipeline

The titler module serves as a small but useful auxiliary component:

- it provides a human-readable summary of the book's thematic identity,
- it aggregates structural scores for quick inspection,
- it offers multiple stylistic variants without requiring manual prompting,

- it integrates seamlessly with the narrator pipeline without influencing the generative process itself.

In summary, the book-title generator is a lightweight, optional module that enhances usability and presentation without compromising the autonomy or deterministic structure of the underlying attractor-based narrative engine.

7 Draft Book Evaluation System

Before a generated draft book enters the narrator pipeline, its structural and stylistic quality is assessed using an automated chapter-level evaluation system. This evaluator provides a quantitative baseline for comparing draft books produced under different training conditions, bias settings, or model variants. The evaluation is intentionally strict: it measures the raw quality of the draft text without moderation, refinement, or narrator intervention.

7.1 Evaluation Prompt

Each chapter is scored using a dedicated LLM prompt designed to assess moderate-level fiction. The prompt defines a clear scoring regime:

- scores range from 1 to 100,
- a score around 50 represents an average moderate-quality chapter,
- scores below 50 are acceptable and common,
- only exceptional chapters should exceed 80.

The evaluator considers seven criteria: content, coherence, readability, originality, creativity, style, and structure. The LLM returns a single line of the form:

$$\text{Score: } \langle \text{number} \rangle$$

without explanation or justification. This ensures consistency and prevents prompt drift.

7.2 Chapter Extraction

The evaluator reads the draft book line by line and extracts chapters based on the presence of chapter markers. Chapters containing fewer than ten sentences are discarded to avoid scoring fragments or malformed sections. Each chapter is represented as a list of raw sentences exactly as produced by the draft generator.

7.3 Double Scoring and Stability

To reduce noise and improve reliability, each chapter is scored twice. The evaluator records:

- the minimum score,
- the maximum score,
- the averaged score (rounded).

This double-scoring mechanism captures small variations in LLM output while remaining computationally lightweight. The averaged score serves as the primary metric for chapter quality.

7.4 Book-Level Aggregation

After all chapters are evaluated, the system computes:

- total number of sentences,
- average minimum score,
- average maximum score,
- average combined score.

These statistics are written to an evaluation file in a compact tabular format:

chapter; sentences; min; max; avg

followed by a final average line summarizing the entire book.

7.5 Role Within the Pipeline

The draft evaluator serves three purposes:

1. It provides a strict baseline for comparing draft books before moderation or narrator refinement.
2. It allows the pipeline operator to select the best-performing draft among multiple candidates.
3. It quantifies the structural and stylistic stability of the underlying attractor model.

Because the evaluator operates on raw draft text, its scores reflect the intrinsic quality of the generative model rather than the smoothing or stabilizing effects of the narrator. This makes it a crucial diagnostic tool for model development, bias modulation experiments, and dataset refinement.

8 Hybrid Narrator Pipelines

The Lightweight Autonomous Narrator Pipeline supports two complementary modes of operation: a *Fully Automated Draft Narrator Pipeline* and a *Fully Automated Iterative Narrator Pipeline*. Both pipelines operate on top of the same GLPG attractor model, optionally augmented with bias modulation, and both produce complete long-form books without human steering. The two variants differ primarily in how chapters are generated and how narrator steering interacts with the attractor dynamics.

8.1 Fully Automated Draft Narrator Pipeline

The draft pipeline processes a pre-generated GLPG book and applies narrator moderation, chapter titling, structural scoring, and inter-chapter transitions. This variant was introduced in earlier work and remains a lightweight and highly effective method for transforming raw attractor output into a coherent literary volume.

Step A: Dataset Selection A human pipeline operator selects one or more books from a modifiable library (e.g. Gutenberg texts or previously generated volumes). These books form the training corpus for the GLPG attractor model.

Step B: Model Training The GLPG model is trained on the selected datasets. An optional bias model may be trained to modulate stylistic or thematic tendencies. Both the base model and the bias model remain compact (60–70 MB) and fully CPU-operational.

Step C: Draft Generation The trained GLPG model generates N autonomous draft books. Each draft is evaluated using structural and stylistic metrics, and the best-performing draft is selected for narrator processing.

Step D: Narrator Pipeline The narrator processes the selected draft chapter by chapter:

- grammatical and semantic moderation using a fixed LLM prompt,
- generation of chapter titles,
- scoring of moderated chapters,
- generation of transitions between consecutive chapters,
- rejection and retry logic based on minimum score thresholds.

The narrator does not alter the attractor-generated structure; it refines, stabilizes, and aligns the draft while preserving its canonical identity.

Step E: Wrap-Up After all chapters are processed:

- a book title is generated from all chapter titles,
- the human operator performs a minimal final review,
- total book scores and statistics are computed,
- the completed book is added to the library.

8.2 Fully Automated Iterative Narrator Pipeline

The iterative narrator pipeline is a new variant introduced in this work. It integrates narrator steering directly into the generative loop, producing chapters *iteratively* rather than moderating a pre-existing draft. This creates a hybrid IE×GR system in which the GLPG attractor model and the LLM-based narrator collaborate during generation rather than sequentially.

Step A: Dataset Selection As in the draft pipeline, datasets are selected from a modifiable library and used to train the GLPG model.

Step B: Model Training The same GLPG model (with optional bias modulation) is used for iterative generation. No architectural changes are required.

Step C: Iterative Generation The narrator pipeline generates the book chapter by chapter, integrating moderation, scoring, and structural alignment directly into the generative loop:

1. The GLPG writer produces a chapter using the current attractor context.
2. The narrator moderates the chapter using the fixed moderation prompt.
3. A chapter title is generated.
4. The chapter is scored; chapters that fall below the threshold are retried.
5. A transition to the previous chapter is generated and validated.
6. A transition score is computed using a dedicated continuity prompt.

7. The attractor context is updated and the next chapter is generated.

This iterative loop continues until the requested number of chapters or lines is reached. The narrator thus becomes an active steering layer that continuously aligns structural, emotional, and thematic flow across the entire book.

If any step fails—either due to insufficient moderation quality, an invalid transition, or a score falling below the required threshold—the entire chapter is retried from the beginning. This mechanism effectively implements a *partial rollback* strategy: the pipeline reverts to the previous stable state and regenerates the chapter until all structural and continuity criteria are satisfied. In practice, this makes the iterative narrator behave like a chapter-level PRE system, ensuring that only coherent, well-aligned chapters advance the global narrative trajectory.

Step D: Wrap-Up The wrap-up phase mirrors the draft pipeline:

- a book title is synthesized from all chapter titles,
- the human operator performs a minimal final review,
- total chapter and transition scores are aggregated,
- the completed book is added to the library.

8.3 Shared Architectural Foundation

Both pipelines rely on the same GLPG attractor model, with or without bias modulation. The narrator layer is identical across both variants, and the LLM prompts used for moderation, titling, scoring, and transitions are shared. The difference lies in the generative flow: the draft pipeline refines a completed attractor draft, while the iterative pipeline co-generates the book in tandem with the attractor system.

Together, these two pipelines form a hybrid architecture capable of producing high-quality autonomous books on lightweight consumer hardware. The draft pipeline offers speed and simplicity; the iterative pipeline offers deeper structural alignment and stronger narrative continuity.

9 Bias Modulation

Bias modulation was introduced in earlier work as an optional steering layer on top of the GLPG attractor system. In the original configuration, the bias MLP was pretrained to approximate the zero function, ensuring that modulation remained negligible unless explicitly scaled. This produced only minimal variation during training, as the modulation function started from a flat baseline.

In the present pipeline, bias modulation is extended with a sinusoidal pretraining regime. Instead of learning $b(h) \approx 0$, the bias MLP is trained to approximate a smooth sinusoidal function of the hidden-state norm:

$$b(h) = \text{offset} + \text{amplitude} \cdot \sin(\text{frequency} \cdot \|h\|).$$

This transforms the bias MLP into a *synthetic mood function*: a compact, deterministic modulation layer whose output varies smoothly with the magnitude of the attractor state. Alpha-scaling then directly controls the intensity of this synthetic mood. Low scaling produces subtle coloration; high scaling induces strong stylistic modulation without destabilizing the underlying dynamics.

9.1 Bias as Internal Cognitive State

Unlike token-level biasing in transformer models, GLPG interprets bias as an *internal state* rather than an external tweak. The attractor dynamics respond to bias as a shift in the direction of the hidden-state trajectory, not as a modification of token probabilities. This distinction is crucial:

- Activation functions are mechanical; modulation functions are cognitive.
- Bias changes the *flow* of the attractor, not its stability.
- GLPG remains stable under extreme mood swings because attractor switching in a compact MLP is exceedingly rare.

The GLPG architecture therefore behaves as a hybrid dynamical system in which bias modulation alters the narrative tone, pacing, and stylistic coloration without compromising canonical structure, lemma routes, or syntactic stability.

9.2 Emergent Styles from the Latent Manifold

The moderator layer does not create style; it stabilizes the emergent worlds produced by the GLPG manifold. All stylistic variability originates from the latent attractor landscape itself. Bias modulation activates different regions of this manifold, producing distinct narrative moods:

- melancholic modulation,
- mythic or epic modulation,
- cosmic or symbolic modulation,
- luminous or moral modulation,
- dark or justice-oriented modulation.

These styles are not injected by the narrator. They emerge from the attractor system and are merely amplified or attenuated by the bias MLP.

9.3 Stability Under Modulation

GLPG remains cognitively stable even under extreme bias scaling. Because the architecture is attractor-based, modulation shifts the direction of the hidden state but does not destabilize the attractor itself. The system continues to produce coherent narrative trajectories, even when the bias norm is large.

This property is unique to compact attractor architectures. Transformer models typically collapse under extreme biasing; GLPG does not.

9.4 Interaction with Entity Tracking

Entity tracking provides a stabilizing semantic backbone that complements bias modulation. While bias activates stylistic attractors, entity tracking anchors the narrative to its core entities, preventing drift and maintaining coherence across long trajectories. Together, these components form a lightweight semantic-cognitive modulation layer:

- bias modulation controls mood and stylistic direction,
- entity tracking stabilizes world identity,
- the moderator ensures structural coherence.

A micro-model equipped with both mechanisms produces emergent narrative worlds that remain coherent, stable, and stylistically expressive.

9.5 Sinusoidal Pretraining

The sinusoidal pretraining regime provides a smooth, interpretable modulation function. Instead of learning arbitrary patterns, the bias MLP learns a controlled oscillatory profile that maps hidden-state magnitude to stylistic intensity. This yields:

- predictable modulation behavior,
- smooth stylistic transitions,
- controllable mood intensity via alpha-scaling,
- stable integration with the attractor dynamics.

The result is a modulation layer that behaves like a synthetic emotional state for the generative system, enabling controlled stylistic variation without compromising structural integrity.

9.6 Alternative Modulation Functions

Although sinusoidal pretraining provides a smooth and interpretable modulation profile, the bias MLP is not restricted to sinus functions. In principle, any continuous mapping from hidden-state magnitude or direction to a modulation vector can be learned. This makes bias modulation an extensible mechanism for expressing a wide range of synthetic internal states.

Possible alternatives include:

- **Polynomial modulation** — smooth monotonic or multi-peak profiles that emphasize specific regions of the attractor manifold.
- **Radial basis modulation** — localized stylistic activation around selected hidden-state norms.
- **Piecewise-linear modulation** — sharp stylistic shifts or controlled transitions between discrete narrative modes.
- **Directional modulation** — functions that depend on the orientation of h rather than its magnitude, enabling thematic steering tied to latent semantic axes.
- **Learned manifold-aware modulation** — training the bias MLP directly on attractor trajectories to produce world-specific or genre-specific internal states.

Because GLPG interprets bias as an internal cognitive state rather than a token-level tweak, these alternative modulation functions would produce distinct narrative moods without destabilizing the attractor dynamics. The architecture therefore supports a broad family of synthetic emotional or stylistic states, of which sinusoidal modulation is only one example.

9.7 Alpha-Scaling as Direct Modulation Strength

In the current implementation, bias modulation is applied through a simple scaling factor α in the forward pass of the bias MLP:

$$b(h) = \alpha \cdot \mathrm{fc}_2(\mathrm{silu}(\mathrm{fc}_1(h))).$$

Although this mechanism was initially introduced as an experimental hack, it proves to be an effective and interpretable control parameter. The value of α directly determines the strength of the modulation applied to the hidden state. In practice, usable values range approximately between -10^{11} and 10^{11}, with 0 disabling the bias, though the exact boundaries are empirical rather than theoretical. Small values

produce subtle stylistic coloration, while large values induce strong mood shifts without destabilizing the attractor dynamics.

Because GLPG interprets bias as an internal cognitive state rather than a token-level tweak, alpha-scaling behaves as a direct intensity control for the synthetic mood generated by the bias MLP. The attractor remains stable even under extreme scaling: modulation changes the *direction* of the hidden state but does not compromise the stability of the attractor itself. This makes alpha-scaling a practical and robust mechanism for exploring a wide range of bias intensities, even in the absence of a formally parameterized modulation layer.

Although currently implemented as a simple multiplicative factor, alpha-scaling could be generalized to more structured modulation schemes in future work, including adaptive scaling, context-dependent scaling, or manifold-aware scaling functions.

10 Training Configuration for Bias Modulation Experiments

To evaluate bias modulation and sinusoidal pretraining within the GLPG architecture, a compact multi-book curriculum was constructed from a curated subset of Project Gutenberg texts. The dataset provides a stable narrative backbone with recurring entities, consistent stylistic structure, and sufficient lexical diversity to support attractor-based world formation.

10.1 Dataset Composition

The training corpus consists of five books, each decomposed into individual sentences with structural annotations:

Sentences	Title	Author
2118	*Alice's Adventures in Wonderland*	Lewis Carroll
1180	*Alice's Adventures Under Ground*	Lewis Carroll
2069	*Alice in Wonderland (One Syllable Edition)*	Carroll / Gorham
2494	*Aesop's Fables*	Aesop
1181	*The Adventures of Reddy Fox*	T. W. Burgess

In total, the curriculum contains:

$$9042 \text{ sentences}, \qquad 130{,}364 \text{ unique samples},$$

yielding an average of:

$$14.4 \text{ samples per sentence}.$$

This density ensures that each sentence is observed under multiple contextual conditions, improving attractor stability and lemma-route consistency.

10.2 Dataset Entities

Entity extraction was performed using grammar-token scanning over nouns, proper names, and plural forms. The resulting statistics are:

$$\text{Total entities} = 2503, \qquad \text{Total lemmas} = 4377.$$

The top 20 entities by occurrence are:

1	alice	881	**11**	day	170
2	fox	419	**12**	king	156
3	time	341	**13**	man	155
4	thing	251	**14**	voice	157
5	reddy	244	**15**	mouse	151
6	way	215	**16**	cat	135
7	head	198	**17**	turtle	121
8	granny	194	**18**	eye	121
9	rabbit	181	**19**	hand	118
10	queen	181	**20**	mock	115

These entities form the semantic backbone of the attractor manifold and are later used by the entity-tracking subsystem to stabilize world identity during generation.

10.3 Model Configuration

Two variants of the GLPG model were trained:

- **Base model**: hidden size 768, model size 59 MB.
- **Bias-enabled model**: hidden size 768 + bias MLP, model size 63 MB.

The bias-enabled model extends the base architecture with a lightweight modulation MLP that supports sinusoidal pretraining and alpha-scaled bias intensity.

10.4 Training Regime

Base Model Training The base attractor model was trained for:

$$300{,}000 \text{ epochs}, \qquad 4{,}318{,}166 \text{ training samples},$$

with a total training time of:

$$5320.8 \text{ seconds}.$$

This establishes the canonical attractor landscape used for all subsequent bias-modulation experiments.

Bias Model Training The bias-enabled model is constructed by extending the fully trained 300,000-epoch base attractor model with a lightweight modulation MLP. Once attached, the bias MLP undergoes a dedicated sinusoidal pretraining stage of 2,000 epochs. This pretraining is applied *only* to the bias MLP: the attractor model remains fixed, and no attractor weights are updated during this phase. The goal is to teach the modulation layer a smooth and interpretable function:

$$b(h) = \text{offset} + \text{amplitude} \cdot \sin(\text{frequency} \cdot \|h\|),$$

which later serves as a synthetic mood function during generation.

After sinusoidal pretraining, the combined model (base attractor + bias MLP) is trained for an additional 100,000 epochs. During this phase, both the attractor and the bias MLP continue learning jointly. This produces a bias-enabled attractor landscape in which modulation interacts naturally with the learned world structure.

In total, the bias-enabled model reaches:

$$400{,}000 \text{ epochs}, \qquad 5{,}756{,}810 \text{ training samples}.$$

The additional 100k epochs—starting from the already trained 300k base model— require:

$$2677.2 \text{ seconds.}$$

This training regime yields a compact attractor model equipped with a sinusoidally pretrained modulation layer, enabling controlled stylistic variation through alpha-scaled bias during generation.

10.5 Purpose of the Training Setup

This training configuration provides:

- a stable attractor manifold with recurring entities,
- a compact semantic backbone for entity tracking,
- a controlled environment for testing sinusoidal bias modulation,
- a lightweight model suitable for consumer-grade hardware,
- and a reproducible baseline for evaluating mood-driven narrative variation.

Together, these components form the experimental foundation for the bias modulation tests described in the following sections.

11 Distilled Model Training Using Generated Books

To investigate how attractor dynamics evolve under second-generation training, a distilled GLPG model was trained on a corpus of eight autonomously generated books. These books were produced by the previously trained base and bias-enabled models under a wide range of modulation intensities. The goal of this experiment is to examine how the attractor manifold sharpens, how entity distributions shift, and how stylistic sensitivity changes when the training data itself originates from the attractor system.

11.1 Generated-Book Corpus

The distilled dataset consists of eight books generated with different bias settings. The corpus composition is summarized below:

Model	Bias	Sentences
base	none	3627
bias	-10^9	2903
bias	-5	3785
bias	0	3781
bias	1	3649
bias	2	3701
bias	10	4032
bias	10^{11}	1640

In total, the generated corpus contains:

$$27118 \text{ sentences}, \qquad 427{,}979 \text{ unique samples},$$

corresponding to:

$$15.8 \text{ samples per sentence.}$$

This density is comparable to the original Gutenberg dataset, but the content is now entirely attractor-generated, making distillation a particularly revealing experiment.

11.2 Entity and Lemma Distribution

Entity extraction over the eight generated books yields:

$$\text{Total entities} = 3848, \qquad \text{Total lemmas} = 7285.$$

The top 20 entities show a clear shift relative to the Gutenberg source data: dominant attractor entities are amplified, while weaker entities fade. This is a typical effect of distillation: the attractor reinforces its own internal semantic cores.

1	alice	5636	**11**	creature	737
2	eye	1324	**12**	tree	662
3	man	1237	**13**	life	624
4	voice	1155	**14**	forest	570
5	day	1074	**15**	world	569
6	reddy	996	**16**	fear	555
7	time	952	**17**	farmer	550
8	heart	941	**18**	air	545
9	sense	806	**19**	fox	534
10	house	755	**20**	way	528

The entity *alice* increases from 881 occurrences in the Gutenberg corpus to 5636 occurrences in the distilled corpus — a factor of 6.4. This illustrates how distillation sharpens the attractor manifold around its dominant semantic centers.

11.3 Distilled Model Training

The distilled model uses the same architecture as the original GLPG model but is trained entirely on the generated books. This produces a second-generation attractor: a dynamical system learning from its own output.

Base Distilled Model

$$\text{Hidden size} = 768, \qquad \text{Model size} = 69 \text{ MB}.$$

Training configuration:

$$900{,}000 \text{ epochs}, \qquad 14{,}202{,}152 \text{ samples},$$

with a total training time of:

$$28524.6 \text{ seconds}.$$

Bias-Enabled Distilled Model The bias-enabled distilled model starts from the base distilled model and performs an additional 300,000 epochs with the bias MLP active:

$$\text{Hidden size} = 768 + \text{bias}, \qquad \text{Model size} = 74 \text{ MB}.$$

Total training:

$$1{,}200{,}000 \text{ epochs}, \qquad 18{,}936{,}443 \text{ samples},$$

with an additional bias-training time of:

$$16927.7 \text{ seconds}.$$

11.4 Purpose and Effect of Distillation

Training on attractor-generated books produces three major effects:

1. **Manifold sharpening** — dominant attractor routes become stronger; weak routes collapse.
2. **Entity amplification** — semantic cores such as *alice*, *man*, *voice*, *heart* are exponentially reinforced.
3. **Enhanced bias sensitivity** — the distilled model responds more strongly and consistently to bias modulation, because the manifold itself has been shaped by bias-generated books.

The result is a second-generation attractor capable of producing richer, more coherent narrative worlds and particularly well suited for further bias-modulation experiments and narrator-pipeline integration.

12 Twelve Generated Books

Twelve books were generated using the GLPG attractor model under a wide range of bias settings, including both first-generation and distilled variants. These books form the artifact collection published within the ecosystem as:

The Moody Parrot: Revisiting the Wonderland Omnibus.

All books were produced with a fixed generation budget of 2000 lines using the draft narrator pipeline. To preserve clarity and page width, the results are split into four compact tables: book titles, generation statistics, evaluation statistics, and emergent moods.

12.1 Book Titles

Model	Bias	Book Title
base	None	Whispers of Wonderland: An Anthology of Mysteries and Adventures
bias	-10^9	Enigmas and Epiphanies: A Journey Through Shadows and Light
bias	-5	Whispers from the Woods: Enigmatic Encounters Along the Path of Life
bias	0	Chronicles of the Unforeseen: A Collection of Enigmatic Journeys
bias	1	Echoes of the Past: A Collection of Unraveled Secrets
bias	2	Enigma's Embrace: A Journey Through the Twists and Turns of the Imagination
bias	10	The Enchanted Realms: Unraveling the Secrets of Alice and Her Extraordinary Companions
bias	10^{11}	Tales from the Wild: Adventures, Alliances, and Allegories
distilled	None	Shadows and Enigmas: A Symphony of Mystery and Resilience
distilled	None	Walking Towards the Unknown: A Journey Through Life's Enigmas
distilled bias	-10	Echoes of the Forgotten: An Enigmatic Adventure Through the Labyrinth of Time
distilled bias	10^6	Whispers of the Forbidden: Exploring the Enigmatic Realms

12.2 Generation Statistics

Model	Bias	Time (s)	Draft Size (KB)	Eval Time (s)	Book Size (KB)	Pipeline Time (s)
base	None	502.7	151	1030.4	287	13571
bias	-10^9	461.7	159	1132.7	231	9672
bias	-5	404.3	153	1074.3	302	12172
bias	0	504.8	153	891.6	295	17011
bias	1	409.4	155	692.2	293	18274
bias	2	409.2	155	993.8	301	11840
bias	10	410.1	155	702.0	321	15355
bias	10^{11}	807.1	174	1088.1	136	11435
distilled	None	790.9	192	505.9	263	26883
distilled	None	590.9	193	551.9	268	25555
distilled bias	-10	606.6	196	409.3	304	17335
distilled bias	10^6	1376.8	218	1395.9	124	18409

12.3 Evaluation Statistics

Draft scores reflect raw chapter quality. Narrator scores reflect moderated chapter quality (minimum score 85). The overall average can be lower because a draft chapter is used when moderation fails.

Model	Bias	Chapters	Avg Lines	Draft Avg	Book Avg
base	None	68	33.2	37.9	86.5
bias	-10^9	50	42.0	38.9	86.1
bias	-5	72	31.4	37.1	86.3
bias	0	68	32.9	37.4	86.1
bias	1	71	31.9	38.4	86.7
bias	2	68	33.4	38.0	86.4
bias	10	71	32.5	38.4	86.4
bias	10^{11}	31	65.4	39.0	83.0
distilled	None	71	32.6	44.5	83.5
distilled	None	68	34.1	46.0	86.3
distilled bias	-10	78	30.0	45.3	86.1
distilled bias	10^6	31	65.2	48.5	86.5

12.4 Moods of Generated Books

Moods arise from the interaction:

$$\text{Mood} = \text{Bias} \times \text{Canon} \times \text{Narrator}.$$

They represent stable attractor directions rather than noise or drift.

Model	Bias	Mood (Core)
base	None	Pastoral Wonder & Time-Mystery (animals, time, myth, farm)
bias	-10^9	Dark Justice Alice (morality, crime, guilt)
bias	-5	Melancholic Quest Alice (emotional journey, wounds, mysteries)
bias	0	Narrative Drift Wonderland (loose scenes, animals, ghosts, dream logic)
bias	1	Epic Mythic Wonderland (kings, artifacts, serpents, quests)
bias	2	Cosmic Symbolic Wonderland (omens, rituals, supernatural causality)
bias	10	Luminous Moral Wonderland (wisdom, compassion, forgiveness)
bias	10^{11}	Hyper-Moral Epic Cosmos (archetypes, cosmic ethics, symbolism)
distilled	None	Mythic Pastoral Peril (village, animals, threat, soft epic)
distilled	None	Mythic Pastoral Peril (stronger canon)
distilled bias	-10	Somber Symbolic Pastoral Drift (farm, animals, abstract entities)
distilled bias	10^6	Somber Mythic Collapse Drift (animals, farm, mythic fragments)

12.5 Summary

The twelve-book study reveals:

- First-generation models: lower raw quality but higher narrative energy.
- Distilled models: higher raw quality but lower narrative energy.
- Bias modulation: stable and predictable mood shifts across the manifold.
- Narrator pipeline: consistently stabilizes all variants at high quality.

These results demonstrate that GLPG attractors form a rich landscape of narrative worlds whose properties can be steered through bias modulation, distillation, and narrator refinement.

13 Lemma Blacklisting

Lemma blacklisting is a deterministic generation-time filtering mechanism that operates directly on the lemma–transition graph of the GLPG model. Each lemma in the vocabulary corresponds to a fixed set of transitions in the attractor manifold: a deterministic mapping that defines which next-lemma candidates are reachable from the current state. This transition map is part of the model's core structure and remains stable after training.

13.1 Deterministic Transition Filtering

During generation, the model proposes a next lemma based on its attractor dynamics and the learned transition graph. Lemma blacklisting intervenes at this precise point:

1. The model computes the next-lemma candidates from the transition map.
2. All candidates whose lemma appears in the blacklist are removed.
3. If the blacklist removes all candidates, the system resamples from the remaining transitions until a valid lemma is found.

Because transitions are deterministic, blacklisting effectively removes entire edges from the lemma graph. A blacklisted lemma cannot be generated: its transitions are filtered out before token emission. The attractor may still *internally* route through states associated with the lemma, but the surface text will never contain it.

13.2 Narrative Control in Mixed-Canon Models

Lemma blacklisting is particularly powerful in mixed-canon settings. When multiple literary worlds coexist inside the same attractor, the model naturally blends protagonists, side characters, and entities from all canons. Without filtering, characters from different books appear together because the attractor treats them as equally valid narrative anchors.

By blacklisting specific lemmas—typically protagonist names or world-specific entities—the pipeline operator can suppress entire canonical identity clusters. This forces the generative system to focus on a single protagonist or world, while still benefiting from the latent stylistic and structural influence of the other canons.

For example, in a tri-canon model combining *Alice in Wonderland*, *Dr. Jekyll and Mr. Hyde*, and *The Metamorphosis*, blacklisting lemmas such as `jekyll`, `hyde`, `utterson`, or `gregor` removes their transitions from the graph. The attractor still expresses their moods, symbolic structures, and thematic dynamics, but explicit references disappear. The result is a purified narrative space in which Alice becomes the sole named protagonist.

13.3 Lightweight Steering Without Retraining

Lemma blacklisting does not modify the model, retrain parameters, or alter the manifold. It simply constrains the visible output by removing transitions from the generation path. This makes it a computationally cheap but highly effective steering tool:

- **No retraining required**: the attractor remains intact.
- **Deterministic effect**: transitions are removed from the graph.
- **Narrative purification**: unwanted entities never appear.
- **Focused storytelling**: a single protagonist or world can be emphasized.

Lemma blacklisting therefore provides precise control over canonical identity and narrative focus while preserving the full expressive richness of the underlying attractor dynamics.

14 World Hybridization

This section evaluates hybrid world formation in a mixed-canon attractor trained on three distinct literary sources: *Alice's Adventures in Wonderland*, *Metamorphosis*, and *The Strange Case of Dr. Jekyll and Mr. Hyde*. The experiment focuses on two mechanisms: (1) lemma blacklisting as a generation-time steering technique, and (2) the difference between draft generation and iterative narrator pipelines.

14.1 Gutenberg Dataset

The hybrid dataset consists of three books with strong and highly distinct canonical identities:

Sentences	Title	Author
2118	Alice's Adventures in Wonderland	Lewis Carroll
995	Metamorphosis	Franz Kafka
1809	The Strange Case of Dr. Jekyll and Mr. Hyde	R. L. Stevenson

Dataset statistics:

$$\text{lemmas} = 4807, \qquad \text{entities} = 2526,$$

$$86642 \text{ unique samples in } 4922 \text{ sentences}, \qquad 17.6 \text{ samples per sentence}.$$

The top 20 entities are:

1	alice	394	11	father	111
2	gregor	296	12	man	112
3	time	190	13	sister	109
4	door	184	14	head	108
5	room	175	15	day	101
6	way	153	16	hyde	97
7	mr	144	17	jekyll	96
8	thing	144	18	mother	90
9	hand	134	19	eye	85
10	utterson	131	20	voice	84

14.2 Base Model Training

The hybrid attractor model was trained with:

$$\text{hidden size} = 768, \qquad \text{model size} = 59 \text{ MB},$$

$$400{,}000 \text{ epochs}, \qquad 7{,}028{,}947 \text{ samples},$$

$$\text{training time} = 8029.8 \text{ s}.$$

This produces a unified manifold in which Alice, Gregor, Jekyll, Hyde, and Utterson coexist as equally accessible narrative anchors.

14.3 Lemma Blacklisting

To focus the hybrid world around Alice, a generation-time lemma blacklist was applied:

`jekyll`, `hyde`, `utterson`, `poole`, `gregor`, `mr`.

Each lemma corresponds to a deterministic set of transitions in the attractor's lemma graph. Blacklisting removes all transitions leading to these lemmas, making them impossible to generate. The attractor still retains their latent structures—mood, tone, symbolic logic—but explicit references disappear. This yields a purified hybrid world in which Alice becomes the sole named protagonist, while the atmospheres of Hyde and Metamorphosis remain present as stylistic undercurrents.

14.4 Draft Narrator Pipeline Books

Three draft books were generated using the standard narrator pipeline:

Time (s)	Size (KB)	Ch.	Eval (s)	Avg Lines	Draft Avg	Book Size (KB)	Narr Time (s)	Book Avg
588.6	165	71	873.6	31.6	38.2	278	13988	86.6
568.3	169	68	683.7	33.4	37.5	271	16397	86.0
633.1	166	66	883.4	34.0	39.3	250	15027	86.1

Draft scores (37–39) reflect raw chapter quality. Narrator scores (86–87) show that the pipeline successfully stabilizes the text into coherent, high-quality chapters.

14.5 Iterative Narrator Pipeline Books

The iterative narrator pipeline enforces chapter-to-chapter continuity. It computes a transition score measuring how well each chapter follows from the previous one. Both the moderation score and transition score threshold are 85.

Time (s)	Size (KB)	Ch.	Avg Score	Transition	Book Title
17840	266	69	86.3	91.3	Whispers from the Veil
15617	237	63	86.1	91.7	Whispers in the Shadows
19020	266	67	86.1	91.9	Enigma's Echoes

Transition scores above 91 indicate strong narrative continuity—significantly higher than draft pipeline books. The iterative pipeline produces smoother, more coherent hybrid narratives.

14.6 Hybridization Effects

The combination of mixed canon, lemma blacklisting, and narrator pipelines produces a distinctive hybrid world:

- Alice becomes the central protagonist.
- Hyde and Metamorphosis contribute atmosphere, tension, and symbolic motifs without explicit entity intrusion.
- Draft pipeline books show strong local coherence but weaker global transitions.
- Iterative pipeline books show high global coherence and stable chapter-to-chapter flow.

Lemma blacklisting proves essential: without it, the hybrid world collapses into a multi-protagonist mixture. With it, the attractor retains the stylistic richness of all three canons while producing a unified Alice-centric narrative.

14.7 Integration into the Ecosystem: The Hybrid Parrot

The six hybridized books produced in this experiment—three generated through the draft narrator pipeline and three through the iterative narrator pipeline—are not standalone artifacts. They form a dedicated component of the broader GLPG ecosystem and are incorporated into the omnibus:

The Hybrid Parrot: Autonomous World Hybridization.

This book serves as the primary demonstration artifact for world hybridization within the narrator architecture. The hybrid world books illustrate how mixed-canon attractors behave under lemma blacklisting, how narrative focus can be steered toward a single protagonist, and how stylistic and symbolic structures from suppressed canons continue to influence the emergent world.

The six books included are:

- **Draft Pipeline Books**
  - *The Veils of Time: Alice's Enigmatic Journey*
  - *Journeys of the Heart and Mind: A Collection of Unfolding Mysteries*
  - *Unraveling the Enigmas: A Collection of Mysteries Unveiled*
- **Iterative Pipeline Books**
  - *Whispers from the Veil: Enigmatic Chronicles of Time and Mystery*
  - *Whispers in the Shadows: Tales of Enigma and Discovery*
  - *Enigma's Echoes: A Symphony of Mysteries, Secrets, and Adventures*

These six artifacts collectively demonstrate the full hybridization workflow:

1. Mixed-canon attractor training (Alice, Hyde, Metamorphosis).
2. Lemma blacklisting to suppress canonical identities from Hyde and Kafka.
3. Draft narrator pipeline for baseline hybrid world formation.
4. Iterative narrator pipeline for high-coherence, chapter-linked hybrid worlds.
5. Integration into the ecosystem as a unified demonstration of autonomous world hybridization.

Within *The Hybrid Parrot*, these books illustrate how attractor dynamics, transition filtering, and narrator refinement interact to produce coherent, Alice-centric hybrid worlds enriched by the latent symbolic structures of Hyde and Metamorphosis. They serve as empirical evidence for the viability of autonomous world hybridization and form the backbone of the book's technical and narrative exposition.

15 Results

This section presents the empirical results of the Lightweight Autonomous Narrator Pipeline across three experimental axes: (1) bias modulation in a single-canon Alice attractor, (2) attractor distillation using autonomously generated books, and (3) mixed-canon world hybridization using lemma blacklisting and two narrator pipelines. In total, the system autonomously generates eighteen complete books, providing a broad empirical basis for evaluating stylistic control, world-model stability, and narrative coherence.

15.1 Bias Modulation in the Alice Attractor

Twelve books were generated using the Alice-only attractor under a wide range of bias intensities. The results show three consistent trends:

1. **Stable stylistic steering.** Bias modulation produces predictable mood shifts across the manifold, ranging from melancholic and mythic to cosmic, luminous, and hyper-moral styles.
2. **High structural stability.** Draft chapters exhibit raw scores around 37–39, while narrator moderation consistently stabilizes chapters at 86–87, independent of bias intensity.
3. **Robust attractor behaviour under extreme modulation.** Even at $\alpha = 10^{11}$, the attractor remains coherent, producing a unified narrative with a distinct stylistic signature.

These results demonstrate that the bias-enabled attractor supports controlled stylistic variation without compromising canonical identity or structural coherence.

15.2 Distillation Effects

Eight autonomously generated books were used to train a second-generation distilled attractor. The results reveal three structural phenomena:

1. **Manifold sharpening.** Dominant entities such as *alice*, *eye*, *voice*, and *heart* increase by factors of 4–6 relative to the Gutenberg corpus, indicating reinforcement of high-probability attractor routes.
2. **Entity amplification and semantic consolidation.** The distilled manifold exhibits stronger canonical centers and reduced low-frequency variation, consistent with entropy compression under self-sampling.
3. **Enhanced bias sensitivity.** Distilled models respond more strongly to modulation, producing sharper and more consistent stylistic moods.

Distillation improves raw draft quality (44–48 average) but reduces narrative energy, confirming that self-sampling drives the attractor toward a smoother, lower-entropy canonical optimum.

15.3 Twelve-Book Corpus Summary

Across the twelve Alice-canon books, the narrator pipeline consistently produces high-quality moderated chapters with average scores between 86.0 and 86.7. Draft quality varies modestly with bias intensity, but moderated quality remains stable due to the fixed moderation prompt and rejection thresholds. Mood classification reveals a rich and predictable stylistic landscape driven by the interaction of bias, canon, and narrator refinement.

15.4 Hybrid World Formation

The mixed-canon attractor trained on Alice, Metamorphosis, and Jekyll/Hyde produces a unified manifold in which all protagonists coexist as equally valid anchors. Lemma blacklisting is essential for steering this manifold: removing `jekyll`, `hyde`, `utterson`, `gregor`, and related lemmas suppresses explicit canonical identities while preserving latent stylistic structures.

15.4.1 Draft Narrator Pipeline

Three hybrid books generated through the draft narrator pipeline exhibit:

- raw draft scores around 38–39,
- moderated scores around 86–87,
- strong local coherence but weaker global transitions.

These books demonstrate that hybrid worlds can be formed reliably, but chapter-to-chapter continuity remains limited without iterative steering.

15.4.2 Iterative Narrator Pipeline

Three additional hybrid books were generated using the iterative narrator pipeline with PRE-like rollback. The results show:

- moderated scores around 86.1–86.3,
- transition scores above 91,
- significantly stronger global coherence,
- smoother chapter-to-chapter flow.

The iterative pipeline consistently outperforms the draft pipeline in global continuity, demonstrating the effectiveness of PRE-style structural validation.

15.5 Hybridization Summary

Across all six hybrid books, the attractor retains the atmospheric, symbolic, and thematic structures of Hyde and Kafka while producing unified Alice-centric worlds. Lemma blacklisting ensures narrative focus, while narrator refinement stabilizes both local and global coherence. The iterative pipeline produces the highest-quality hybrid worlds, confirming that structural validation and rollback are essential for mixed-canon generation.

15.6 Overall Findings

The eighteen-book study reveals four overarching results:

1. **Architectural stability.** Compact attractor models remain coherent under extreme modulation, mixed canons, and distillation.
2. **Modular controllability.** Bias modulation, entity tracking, lemma blacklisting, and narrator refinement provide interpretable and predictable steering mechanisms.
3. **Narrative coherence.** The narrator pipeline consistently stabilizes raw attractor output, with the iterative variant producing strong global continuity.
4. **Ecosystem viability.** The pipeline forms a reproducible, inspectable generative ecosystem capable of producing diverse, coherent, and stylistically controlled long-form books.

These results collectively demonstrate that long-form narrative generation, stylistic modulation, and world hybridization can emerge from lightweight, architecturally grounded systems rather than large-scale probabilistic models.

16 Limitations

Although the Lightweight Autonomous Narrator Pipeline demonstrates that coherent long-form narrative generation can emerge from compact attractor architectures, several limitations remain. These limitations reflect both structural properties of the pipeline and broader constraints inherent to lightweight generative systems.

16.1 Dependence on the Moderation LLM

The final narrative quality depends strongly on the capabilities of the moderation LLM used in the narrator layer. Larger models provide more reliable semantic smoothing, stronger stylistic alignment, and more consistent chapter coherence. While the attractor produces structurally stable drafts, the narrator remains the primary source of linguistic refinement. As a result, moderation quality scales with LLM size: lightweight moderators are sufficient for structural stability but cannot match the stylistic richness of frontier-scale models.

16.2 Need for World-Specific Moderation Prompts

The pipeline currently uses generic moderation prompts designed to operate across all canons and stylistic domains. Although this preserves autonomy and avoids manual steering, it also limits the achievable quality. In practice, high-quality narrative refinement often requires world-specific or canon-specific hints—much like an editor guiding an author. The present work intentionally avoids such domain tuning to demonstrate full autonomy, but this choice constrains the upper bound of stylistic fidelity.

16.3 Continuity Limitations in Long-Range Structure

The iterative narrator pipeline significantly improves chapter-to-chapter continuity, yet long-range narrative planning remains limited. The attractor generates emergent trajectories rather than globally planned story arcs, leading to narratives that resemble dream-like chains of events or improvisational sequences rather than tightly plotted novels. Achieving long-range structural planning would require additional architectural components capable of operating at larger temporal scales.

16.4 Emergent and Unpredictable Narrative Outcomes

Because the attractor manifold is emergent and deterministic only at the local level, the global narrative trajectory cannot be predicted without generating and reading the entire book. Each run explores a different region of the manifold, and stylistic or thematic outcomes vary accordingly. Although an LLM can rapidly summarize the generated text, the system does not provide built-in mechanisms for predicting or constraining high-level narrative outcomes.

16.5 Lightweight System Constraints

The pipeline is designed for consumer-grade hardware and compact models. Within this domain, it likely represents state-of-the-art capability: coherent multi-chapter books, controllable stylistic variation, and mixed-canon hybridization are rare achievements for systems of this scale. However, the pipeline is not intended to compete with frontier-scale datacenter models or agentic pipelines operating over massive compute budgets. Its purpose is architectural demonstration rather than frontier performance.

16.6 Narrator Pipeline as Ecosystem Artifact

The narrator pipeline serves primarily as an ecosystem artifact within the *Triadic Cosmos* framework. Its goal is to illustrate principles of architectural general intelligence—modularity, interpretability, safety, structural determinism—rather than to function as a fully autonomous, publication-ready author. The

generated books are valuable as emergent artifacts and empirical demonstrations, but they are not intended to replace human-written literature or professional editorial workflows.

16.7 Absence of High-Level Authorial Intent

Finally, the system lacks mechanisms for expressing high-level authorial intent, themes, or planned narrative arcs. All books are fully emergent: the operator cannot specify plot goals, emotional trajectories, or symbolic structures beyond the implicit biases encoded in the attractor manifold. This limitation is structural and reflects the pipeline's commitment to autonomy rather than scripted generation.

In summary, the narrator pipeline demonstrates strong architectural capabilities within a lightweight setting, but its performance remains bounded by moderation quality, lack of long-range planning, and the inherent unpredictability of emergent attractor dynamics.

17 Conclusion

The Lightweight Autonomous Narrator Pipeline demonstrates that long-form narrative generation, stylistic steering, and mixed-canon world hybridization can emerge from compact attractor architectures operating entirely on consumer-grade hardware. Rather than relying on large transformer models or extended-context attention, the pipeline integrates deterministic GLPG attractor dynamics with modular cognitive layers: a narrator for structural refinement, a sinusoidally pretrained bias module for synthetic mood control, and a high-bandwidth entity-tracking vector for world stabilization. Together, these components form a hybrid IE×GR system in which narrative competence arises from architectural structure rather than brute-force scale.

Across three experimental axes—bias modulation, attractor distillation, and mixed-canon hybridization—the system autonomously generates eighteen complete books. The results show predictable stylistic variation under bias scaling, manifold sharpening and entity amplification under distillation, and coherent Alice-centric hybrid worlds produced through lemma blacklisting and narrator refinement. The iterative narrator pipeline further demonstrates that PRE-like rollback mechanisms can enforce strong chapter-to-chapter continuity, yielding high-coherence narratives that exceed the structural stability of draft-only generation.

These findings highlight a broader architectural principle: generative capabilities traditionally associated with large-scale models can emerge from small, interpretable systems when narrative structure is embedded directly into the attractor manifold and reinforced through modular cognitive layers. The pipeline therefore serves as a domain-bound demonstration of *architectural general intelligence*, illustrating how hybrid intelligent systems can be built from compositional components that remain inspectable, reproducible, and computationally lightweight.

Finally, the experiments reveal structural risks inherent to self-sampling ecosystems—entropy compression, fixed-point convergence, and stylistic homogenization—and underscore the continued importance of human creativity as a source of high-entropy boundary conditions. Within the broader *Triadic Cosmos* ecosystem, the narrator pipeline provides both a practical generative engine and a conceptual reference point for studying modular, architecturally grounded approaches to artificial intelligence.

A A Theoretical Perspective on AI Overshoot Dynamics

A.1 Low-Entropy Convergence Under Self-Sampling

The experiments in this paper reveal a structural phenomenon also observed in distilled models: when an attractor is repeatedly trained on its own output, the system begins to drift toward the moderator boundary. Draft scores increase because the manifold becomes smoother, narrower, and more aligned with the moderator's expectations. However, moderated scores do not automatically rise. Instead, moderation becomes more difficult: the narrator must work harder to inject variation, tension, and narrative energy into an increasingly homogenized, low-entropy attractor.

This behaviour reflects a general statistical principle: any system that repeatedly samples from its own states tends to lose entropy. In the context of AI ecosystems, this manifests as a progressive collapse of variation, outliers, and creative attractors. The manifold converges toward a minimum-variance equilibrium where stylistic diversity is replaced by canonical smoothness.

From a theoretical standpoint, this collapse can be described through three interacting phenomena:

1. **Iterated Distribution Collapse** — repeated sampling from the model's own output causes the empirical distribution to contract, eliminating rare events and amplifying dominant modes.

2. **Entropy Compression Under Self-Sampling** — each generation of self-produced data reduces entropy, because the model preferentially reproduces high-probability structures while suppressing low-probability attractors.

3. **Markov Fixed-Point Convergence** — the attractor behaves as a Markov process whose repeated application drives the system toward a fixed point: a stable, low-variance canonical optimum.

As AI-generated content increasingly permeates the internet, code repositories, image datasets, and video corpora, the ratio of human to AI data shifts generation by generation. Scaling laws demand ever-larger datasets, but the available "new" human data grows slowly. The result is a structural overshoot: models ingest more AI-generated data, which accelerates entropy compression and pushes the ecosystem toward homogenization.

This leads to two long-term consequences:

- **Frontier convergence** — all large-scale models drift toward the same average-quality, low-entropy optimum. Qualitative differentiation disappears; only inference speed and cost remain as competitive axes.

- **Overshoot dynamics** — as qualitative gains diminish, scaling requires disproportionate increases in compute, energy, labour, and infrastructure. Returns shrink while costs escalate, producing multi-domain overshoot: compute overshoot, energy overshoot, software overshoot, labour overshoot, and infrastructure overshoot.

This transformation shifts AI from an innovative high-tech frontier to a commodity market dominated by cost efficiency rather than quality. The collapse loop is already visible in open models, which often share similar training data, similar architectural patterns, and similar performance profiles. Variations between them increasingly reflect engineering optimizations rather than fundamental differences in generative capability.

In summary, self-sampling drives attractors toward a narrow canonical optimum. Draft quality rises because the manifold becomes smoother, but moderated quality plateaus or declines because narrative energy and structural variation diminish. This phenomenon scales beyond individual models: as AI-generated data becomes dominant in global datasets, the entire ecosystem risks converging toward a low-entropy fixed point, triggering overshoot dynamics across all supporting domains.

A.2 Human Creativity as a Safeguard Against Low-Entropy Convergence

The collapse dynamics described above highlight a structural risk inherent to self-sampling AI ecosystems: when a model repeatedly trains on its own output, the attractor manifold contracts, entropy decreases, and the system converges toward a narrow canonical optimum. Draft scores rise because the manifold becomes smoother and more predictable, but moderated scores plateau or decline because narrative energy, structural variation, and creative tension diminish.

This phenomenon is not limited to literary attractors. As AI-generated content increasingly permeates global datasets—text, code, images, audio, and video—the ratio of human to AI data shifts generation by generation. Scaling laws demand ever-larger datasets, but the supply of new human-created material grows slowly. The result is a structural overshoot: more AI data is injected into the training pipeline, accelerating entropy compression and pushing the ecosystem toward homogenization.

The long-term consequence is a convergence toward a low-entropy fixed point where all frontier models exhibit similar average quality, similar stylistic profiles, and similar canonical smoothness. Qualitative differentiation disappears, and competition shifts toward inference speed, cost, and infrastructure efficiency. This transition transforms AI from an innovative high-tech frontier into a commodity market, triggering overshoot dynamics across compute, energy, labour, and software ecosystems.

Yet the hybridization experiments reveal an important counterbalance: human creativity remains a critical driver of entropy injection. Human-written text, human-designed worlds, human narrative structures, and human conceptual diversity introduce irregularity, asymmetry, and surprise—properties that self-sampling systems cannot generate on their own. Even in the age of large-scale autonomous generation, human creativity provides the high-entropy boundary conditions that prevent attractors from collapsing into sameness.

In this sense, human creativity acts as a structural safeguard:

- It injects non-Markovian variation into the dataset.
- It prevents fixed-point convergence by introducing new attractor basins.
- It maintains narrative, stylistic, and conceptual diversity.
- It counteracts entropy compression caused by self-sampling.

Thus, even as AI ecosystems grow in scale and autonomy, their long-term evolution remains fundamentally dependent on human creative input. Without continuous human-generated novelty, the system drifts toward a low-entropy canonical optimum; with it, the manifold remains open, diverse, and capable of producing new worlds rather than merely refining old ones.

This demonstrates that—even in the age of AI—human creativity is not merely valuable but structurally necessary for the continued evolution of generative systems.

B Triadic Cosmos Ecosystem Context

The present paper is part of the broader *Triadic Cosmos* ecosystem: a unified research program that develops a compositional ontology of intelligence together with modular, inspectable computational architectures. The ecosystem spans conceptual theory, executable reference implementations, and fully documented empirical demonstrations, forming a coherent framework for studying hybrid intelligent systems.

All materials, extended notes, and evolving documentation are publicly available in the open GitHub project:

`https://github.com/triadic-cosmos`

Within the `triadic-system` repository, a complete reference implementation of the DMLG and GLPG engines, the narrator pipeline, and all supporting modules is provided under the AGPL-3.0 license.

References

[1] J. De Witte, “The GLP–Graph Book Authoring Pipeline,” Triadic Cosmos Ecosystem, 2026.

[2] J. De Witte, “A Narrator Pipeline driven by DMLG,” Triadic Cosmos Ecosystem, 2026.